

~~Which exchange-correlation functional should I use?~~

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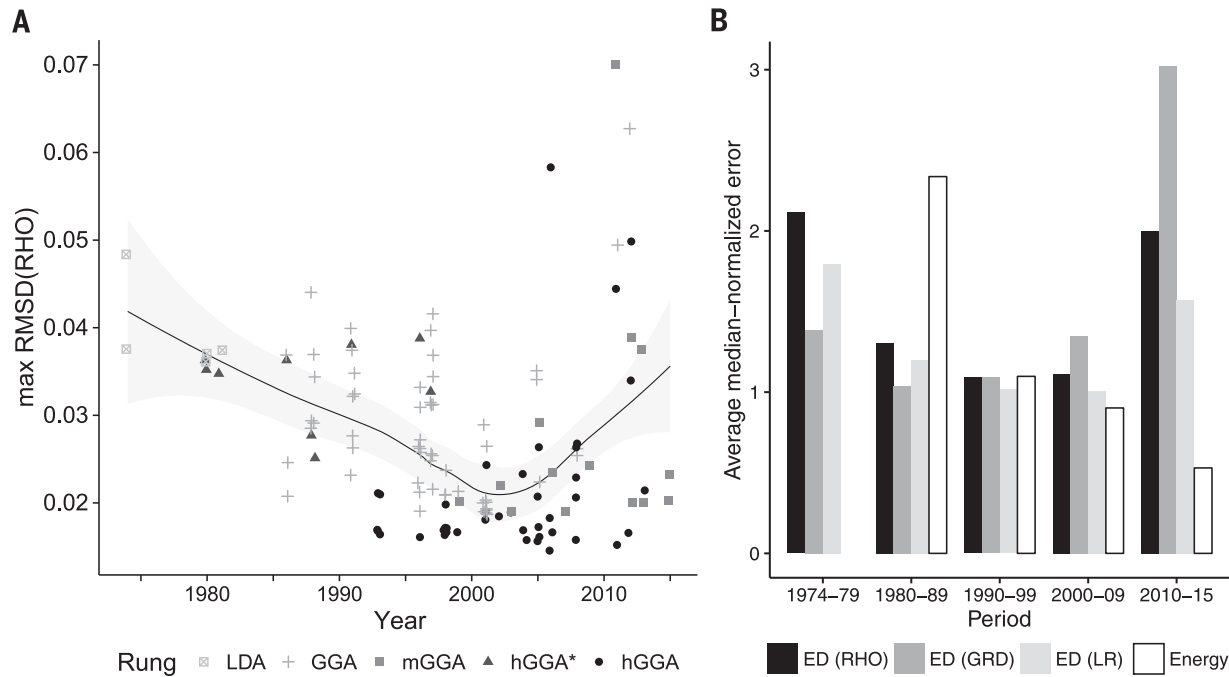
CASTEP Training Workshop 2025

A guide to exchange-correlation functionals

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CASTEP Training Workshop 2025

Exchange-correlation functionals: A zoo



Science 355, 49

Table 1. The L1 methods (yielding the best densities), their rungs, years, and normalized errors (NE).

Method	Rung	Year	Max NE
CCSD	Ab initio		0.000
MP4sdq	Ab initio		0.246
MP3	Ab initio		0.967
MP2	Ab initio		1.514
mPW3PBE	hGGA	1998	1.778
APFD	hGGA	2012	1.813
B3PW91	hGGA	1993	1.816
PBE0	hGGA	1999	1.818
B98	hGGA	1998	1.826
BHHLYP	hGGA	1993	1.851
B97-3	hGGA	2005	1.883
mPW1PBE	hGGA	1998	1.910
B3P86	hGGA	1993	1.937
O3LYP	hGGA	2001	1.947
PBE1KCIS	hGGA	2005	1.954
mPW1PW91	hGGA	1998	1.955
B97-1	hGGA	1998	1.962
HSE06	hGGA	2006	1.982
PBEh1PBE	hGGA	1998	1.983
B97-2	hGGA	2001	2.018
B1B95	hGGA	1996	2.033
TPSS	mGGA	2003	2.042
TPSSh	hGGA	2003	2.045
TPSSm	mGGA	2007	2.077
X3LYP	hGGA	2005	2.084
SCAN	mGGA	2015	2.107
B3LYP	hGGA	1993	2.123

Table 2. The L2 methods (yielding the worst densities), their rungs, years, and normalized errors (NE).

Method	Rung	Year	Max NE
$X\alpha^*$	LDA	1974	3.777
SP86	GGA	1986	3.821
M06-L	mGGA	2006	3.974
SVWN1RPA	LDA	1980	3.977
SPBE	GGA	1997	3.978
SVWN	LDA	1980	3.984
SPZ81	LDA	1981	3.985
SPW91	GGA	1991	3.989
M06-2X	hGGA	2006	4.027
SOP	GGA	1997	4.182
SLYP	GGA	1988	4.429
M08-SO	hGGA	2008	4.676
SLATER*	LDA	1974	4.864
M08-HX	hGGA	2008	4.880
SOGGA11	GGA	2011	4.971
M06	hGGA	2006	5.420
M06-HF	hGGA	2006	6.125
N12	GGA	2012	6.709
N12-SX	hGGA	2012	6.970
M05	hGGA	2005	7.652
MN12-L	mGGA	2012	8.995
M11	hGGA	2011	10.191
MN12-SX	hGGA	2012	13.005
M11-L	mGGA	2011	15.316

* $\alpha = 0.7$ for $X\alpha$ and $\alpha = 2/3$ for SLATER (see supplementary materials for references).

Results for the 2019 edition of the Annual DFT Popularity Poll: Primera Divisió

functional	like	neutral	hate	blank	none	#votes	points	average						
1 PBE	180	44.1%	49	11.4%	28	7.1%	10	2.1%	276	81.3%	836	3.0		
2 PBE0	124	41.0%	77	24.8%	29	7.1%	30	8.1%	269	80.6%	428	-4.4		
3 HSE	112	35.4%	89	26.2%	18	5.6%	62	17.1%	272	81.3%	498	-3.7		
4 PBE-D	107	35.5%	78	22.2%	27	8.1%	60	16.8%	262	81.9%	372	-3.9		
5 B3LYP	99	40.0%	114	46.8%	49	19.4%	34	11.4%	256	80.6%	380	-4.1		
6 PBEsol	97	33.4%	85	27.8%	26	8.4%	48	10.1%	266	81.9%	391	-3.7		
7 LDA	102	43.3%	87	32.2%	28	11.1%	22	10.1%	278	90.6%	385	-1.0		
8 SCAN	78									80.6%	394	-4.0		
9 PW91	61									80.6%	268	0.88		
10 B3LYP-D	50									90.6%	248	0.60		
11 wB97X-D	56									80.6%	246	0.80		
12 CAM-B3LYP	48									81.3%	205	0.79		
13 revPBE	46									80.6%	210	0.79		
14 B97-D	34									80.6%	212	0.7		
15 M06-2X	50									81.3%	185	0.6		
16 wB97X-V	31									80.6%	159	0.60		
17 TPSSh	25	4.1%	102	32.8%	28	8.4%	96	28.1%	44	4.8%	254	80.7%	149	0.38
18 M06	38	11.1%	83	19.3%	68	16.5%	76	18.1%	33	11.1%	266	81.9%	138	0.47
19 revTPSS	17	4.1%	105	32.8%	27	6.1%	102	30.1%	45	4.4%	252	80.6%	129	0.44
20 revTPSS-D	19	4.1%	99	32.8%	31	18.4%	108	30.1%	44	14.8%	254	80.7%	122	0.4

total number of participants: 288

Primera divisió &
Segona divisió

Ab initio?

libxc: over 700
functionals

<http://www.marcelswart.eu/dft-poll>

Density Functional Theory

Many-body Schrödinger equation (the “truth”):



$$\hat{H}\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$$

Hohenberg-Kohn theorems



Mapping from a many-body problem to a one-body problem

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \rightarrow n(\mathbf{r})$$

$$E[n] = F[n] + \int d\mathbf{r} V(\mathbf{r})n(\mathbf{r}) = T[n] + U[n] + \int d\mathbf{r} V_{\text{ext}}(\mathbf{r})n(\mathbf{r})$$

Kohn-Sham formulation



Model the density with non-interacting single-electron orbitals

$$E[n] = -\frac{1}{2} \sum_{i=1}^N \nabla^2 \psi_i + \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + E_{\text{xc}}[n] + \int d\mathbf{r} V_{\text{ext}}(\mathbf{r})n(\mathbf{r}) \quad n(\mathbf{r}) = \sum_{i=1}^N |\psi_i(\mathbf{r})|^2$$

Density Functional Theory



Kohn-Sham formulation

$$E[n] = -\frac{1}{2} \sum_{i=1}^N \nabla^2 \psi_i + \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \int d\mathbf{r} V_{\text{ext}}(\mathbf{r})n(\mathbf{r}) + E_{\text{xc}}[n] \quad n(\mathbf{r}) = \sum_{i=1}^N |\psi_i(\mathbf{r})|^2$$

Kinetic energy of the
non-interacting particles

$$T_s[n] = -\frac{1}{2} \sum_{i=1}^N \nabla^2 \psi_i \quad \text{vs.} \quad T[n]$$

Hartree-term: Coulombic
interaction of electrons

$$E_H[n] = \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \quad \text{vs.} \quad U[n]$$

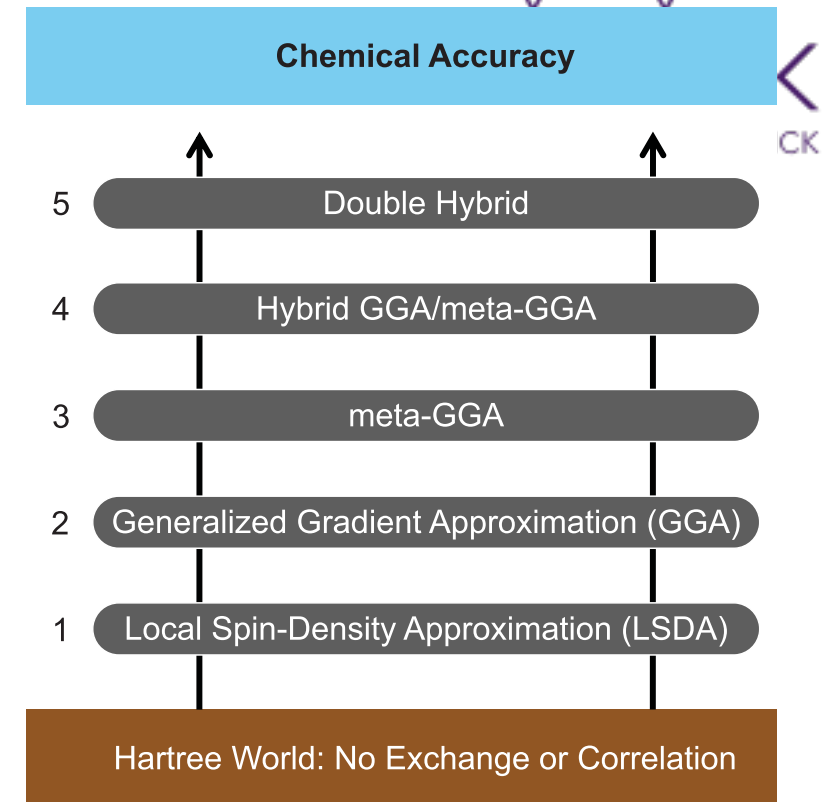
Exchange-correlation functional has to
model:

- kinetic energy of the many-body system
- non-Coulombic interactions

$$E_{\text{xc}}[n] = \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' n(\mathbf{r}) \frac{n(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

Exchange-correlation functionals

- The form of the exchange-correlation functional is unknown
- There exist a series of approximations: “Jacob’s ladder” to chemical accuracy
- Chemical accuracy: thermochemical energetics (1 kcal/mol error in formation energies), molecular geometries
- Errors in DFT can be attributed to the limitation of XC functionals
 - Self-interaction error (electron self-repulsion) c.f. hydrogen
 - Derivative discontinuity
 - Strongly correlated systems
 - Non-local correlation - weak interactions



from N Mardirossian and M Head Gordon,
Mol. Phys. **115** 2315

$$E_H[n] = \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

$$E(N + \delta) = (1 - \delta)E(N) + \delta E(N + 1)$$

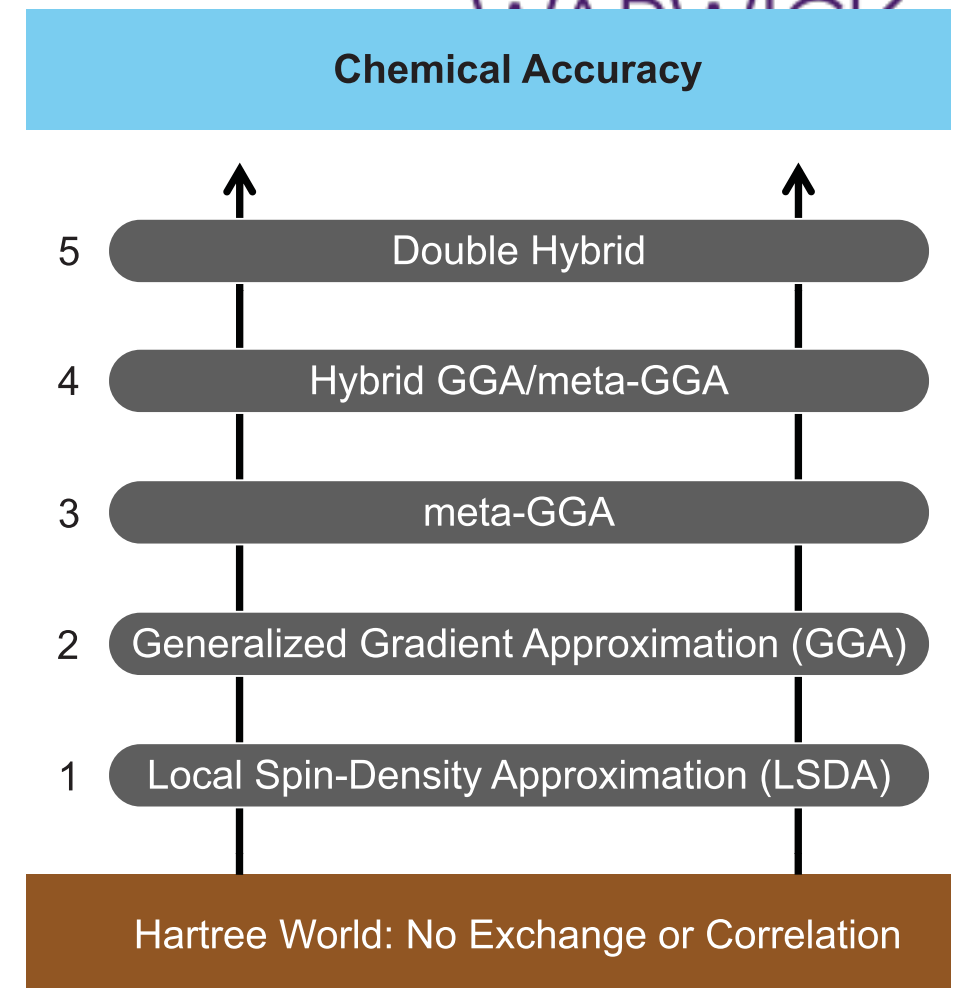
Exchange-correlation functionals

(m)GGA + exact exchange + MP2 correlation

(m)GGA + exact exchange

semi-local functionals

$$\left\{ \begin{array}{l} E_{xc} = \int d\mathbf{r} \, \varepsilon_{mGGA}(n(\mathbf{r}), \nabla n(\mathbf{r}), \nabla^2 n(\mathbf{r}), \tau(\mathbf{r}))n(\mathbf{r}) \\ E_{xc} = \int d\mathbf{r} \, \varepsilon_{GGA}(n(\mathbf{r}), \nabla n(\mathbf{r}))n(\mathbf{r}) \\ E_{xc} = \int d\mathbf{r} \, \varepsilon_{LDA}(n(\mathbf{r}))n(\mathbf{r}) \end{array} \right.$$



from N Mardirossian and M Head Gordon,
Mol. Phys. **115** 2315

ε_{LDA} , ε_{GGA} and ε_{mGGA} are *unknown*, need fitting!

Generalities



- All XC functionals are approximations
 - DFT (the theory) is exact
 - Reports of 'Failures of DFT' are actually reports of the failure of the XC functional (approximation)
- No functional (so far) is accurate for all properties of interest
 - No matter what functional is invented, someone will always find a case where it fails
- Any functional can be applied to any electronic structure problem
 - But they have *known* limitations
 - It is still *ab initio*, but we use experience, benchmarks and intuition to decide which ones to use

Local Density Approximation

Typical errors:

- Binding energies too large
- Bonds and lattice parameters too small
- Incorrect order of phase stability
- Error in energetics in magnetic materials

$$E_{xc} = \int d\mathbf{r} \, \varepsilon_{LDA}(n(\mathbf{r}))n(\mathbf{r})$$

Curtiss *et al.*: Computation of enthalpies of formation

1073

TABLE IV. Deviation of enthalpies calculated by DFT methods with experiment.^a

Molecule	LDA	Deviation (Expt.-Theory)					
		BLYP	BP86	BPW91	B3LYP	B3P86	B3PW91
<i>G2-1 test set</i>							
LiH	3.3	0.1	0.5	−4.6	0.4	1.2	−3.3
BeH	11.2	7.3	8.5	6.5	8.2	10.0	7.4
CH	9.9	1.8	4.1	−0.1	1.7	3.9	0.0
CH ₂ (³ B ₁)	24.1	0.0	8.5	2.8	2.1	10.3	4.2
CH ₂ (¹ A ₁)	20.7	−0.6	4.2	−3.7	0.2	4.9	−2.7
CH ₃	35.2	0.4	10.4	0.3	3.3	13.1	2.9
CH ₄	46.8	−2.3	10.4	−3.4	1.6	14.1	0.2
NH	13.7	6.0	8.6	3.8	4.6	7.1	2.5
NH ₂	29.7	8.0	13.5	4.3	6.5	11.9	3.0
NH ₃	44.3	4.3	13.3	0.3	3.5	12.4	−0.4
OH	19.5	3.4	6.6	2.0	1.8	5.1	0.4
OH ₂	37.3	0.6	7.8	−0.9	−1.3	5.8	−3.1
FH	22.9	0.4	4.1	−0.2	−1.6	2.2	−2.3

L. A. Curtiss *et al.*, *JCP* **106**, 1063

LDA parameterisations

$$E_{xc} = \int d\mathbf{r} \, \varepsilon_{\text{LDA}}(n(\mathbf{r}))n(\mathbf{r})$$

XC_FUNCTIONAL : LDA

- Exchange: Slater exchange
- Correlation: parameterisations based on accurate calculations of the Homogenous Electron Gas
 - Vosko-Wilk-Nusair 5 (VWN5): Can. J. Phys. 58, 1200 (1980)
 - Perdew-Zunger: PRB 23, 5048 (1981)
 - Perdew-Wang: PRB 45, 13244 (1992)
 - VWN3: Can. J. Phys. 58, 1200 (1980)

$$\varepsilon_{\text{LDA},x} \propto n^{1/3}$$

Generalised Gradient Approximations

$$E_{xc} = \int d\mathbf{r} \, \varepsilon_{GGA}(n(\mathbf{r}), \nabla n(\mathbf{r}))n(\mathbf{r})$$

- More flexible functional form
- Semi-local: information about nearby regions via the gradient
- Not very successful before 1996 (PBE: Generalised Gradient Approximation Made Simple)
- Theoretical limits and constraints explicitly built in
- Modifies LDA exchange and correlations using “enhancement factors”

Generalised Gradient Approximations

Lattice constants (Å)

Binding energies (eV / atom)

Solids	LSDA	PBE	PBEsol	Expt.	LSDA	PBE	PBEsol	Expt.
Li	3.362	3.431	3.426	3.453	1.810	1.605	1.677	1.658
Na	4.051	4.198	4.170	4.214	1.256	1.082	1.154	1.119
Ca	5.332	5.518	5.448	5.553	2.220	1.917	2.117	1.86
Sr	5.791	6.027	5.916	6.045	1.893	1.609	1.808	1.73
Ba	4.770	5.030	4.894	4.995	2.246	1.871	2.109	1.91
Al	3.983	4.035	4.011	4.018	4.038	3.438	3.817	3.431
Cu	3.522	3.633	3.565	3.595	4.545	3.474	4.027	3.524
Rh	3.759	3.831	3.781	3.794	7.563	5.688	6.642	5.783
Pd	3.844	3.942	3.876	3.876	5.016	3.714	4.435	3.938
Ag	4.002	4.145	4.050	4.062	3.642	2.516	3.078	2.985
C	3.533	3.569	3.552	3.553	9.011	7.714	8.275	7.545
SiC	4.332	4.378	4.355	4.346	7.457	6.401	6.876	6.478
Si	5.405	5.468	5.432	5.421	5.348	4.559	4.940	4.685
Ge	5.631	5.768	5.680	5.644	4.628	3.716	4.144	3.918
GaAs	5.615	5.752	5.665	5.640	4.095	3.148	3.555	3.337
LiF	3.913	4.070	4.007	3.972	4.945	4.322	4.474	4.457
LiCl	4.968	5.151	5.063	5.070	3.835	3.364	3.518	3.586
NaF	4.506	4.705	4.632	4.582	4.384	3.826	3.959	3.970
NaCl	5.467	5.695	5.603	5.569	3.503	3.097	3.223	3.337
MgO	4.170	4.261	4.223	4.189	5.863	4.973	5.299	5.203
ME	-0.082	0.051	-0.012		0.642	-0.121	0.234	
MAE	0.082	0.060	0.035		0.642	0.144	0.253	
MRE(%)	-1.73	1.10	-0.24		16.50	-3.68	5.97	
MARE(%)	1.73	1.29	0.73		16.50	4.23	6.52	

GGA parameterisations

$$E_{xc} = \int d\mathbf{r} \, \varepsilon_{GGA}(n(\mathbf{r}), \nabla n(\mathbf{r}))n(\mathbf{r})$$

XC_FUNCTIONAL : PW91

- J. P. Perdew and Y. Wang, *PRB* **45** 13244 (1992)

XC_FUNCTIONAL : PBE

- J. P. Perdew, K. Burke and M Ernzerhof, *PRL* **77** 3865 (1996)

XC_FUNCTIONAL : RPBE

- B. Hammer, L. B. Hansen and J. K. Norskov, *PRB* **59** 7413 (1999)

XC_FUNCTIONAL : WC

- Z. Wu and R. E. Cohen, *PRB* **73**, 235116 (2006)

Meta-Generalised Gradient Approximations



$$E_{\text{xc}} = \int d\mathbf{r} \, \varepsilon_{\text{mGGA}}(n(\mathbf{r}), \nabla n(\mathbf{r}), \nabla^2 n(\mathbf{r}), \tau(\mathbf{r})) n(\mathbf{r})$$
$$\tau(\mathbf{r}) = \frac{1}{2} \sum_{i=1}^N |\nabla \psi(\mathbf{r})|^2 \quad V_{\text{xc}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}}{\delta n} = V_{\text{GGA}}(\mathbf{r}) + \hat{V}_{\tau}(\mathbf{r})$$

- Even more flexible functional form
- Still semi-local
- Not very successful before 2015 (SCAN: Strongly Constrained and Appropriately Normed functional)
- More theoretical limits, constraints and accurate cases explicitly built in
- Kinetic energy density: solve the Generalised Kohn-Sham equations (XC potential not multiplicative)
- Improved energetics and structural properties
- Band gaps closer to experimental ones

mGGA parameterisations

$$E_{xc} = \int d\mathbf{r} \, \varepsilon_{mGGA}(n(\mathbf{r}), \nabla n(\mathbf{r}), \nabla^2 n(\mathbf{r}), \tau(\mathbf{r})) n(\mathbf{r})$$

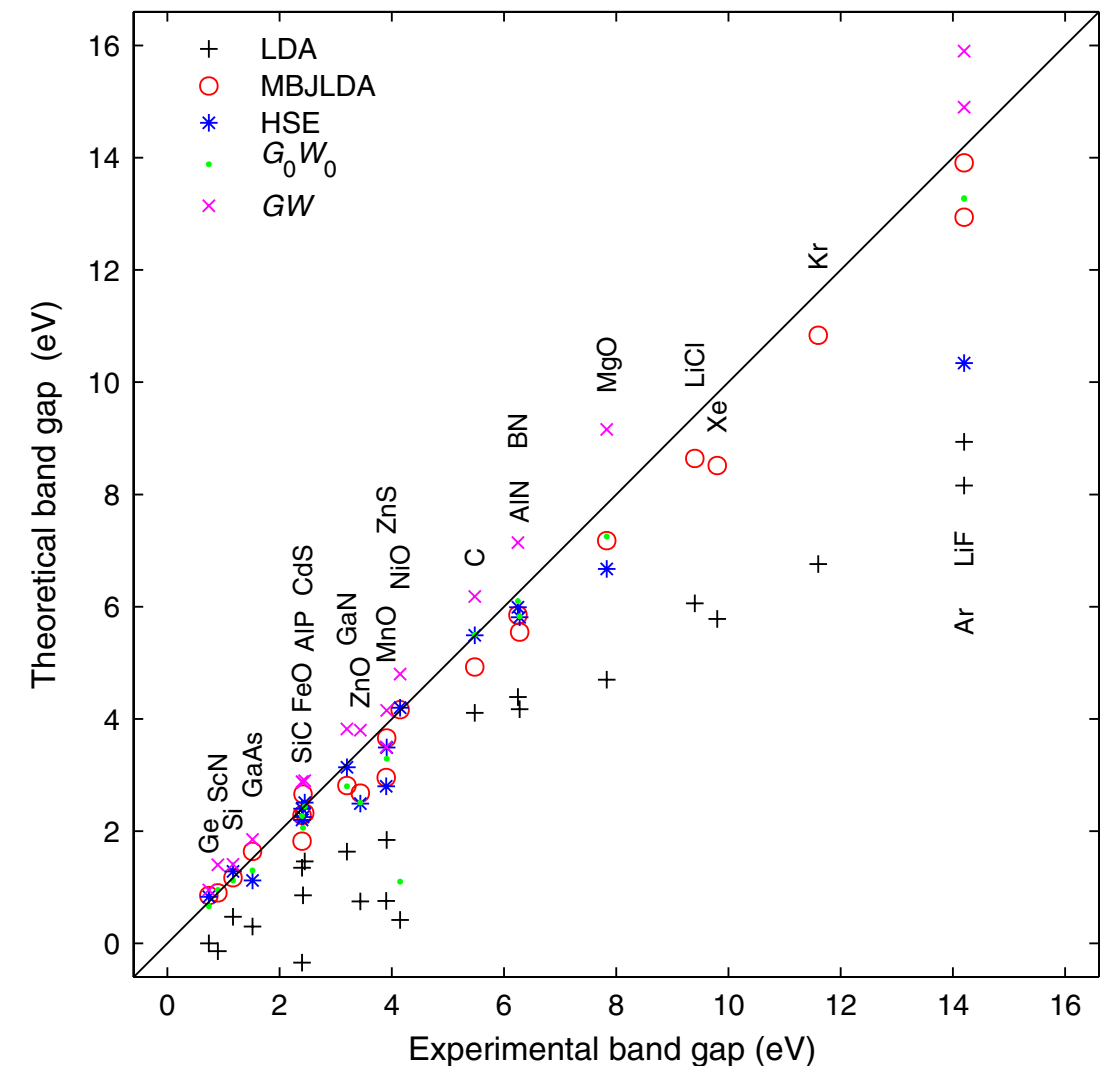
$$V_{mBJ}(\mathbf{r}) = V(n(\mathbf{r}), |\nabla n(\mathbf{r})|, \tau(\mathbf{r}))$$

$$\tau(\mathbf{r}) = \frac{1}{2} \sum_{i=1}^N |\nabla \psi(\mathbf{r})|^2$$

$$V_{mGGA}(\mathbf{r}) = \frac{\delta E_{xc}}{\delta n} = V_{GGA}(\mathbf{r}) + \hat{V}_\tau(\mathbf{r})$$

PRL 102, 226401

- Modified Becke-Johnson (Tran-Blaha) *potential*:
Tran and Blaha, PRL 102, 226401 (2009)
 - *potential only* mGGA, **no** total energy, **no** forces etc.
 - improves band gaps
- TPSS mGGA: Tao, Perdew, Staroverov, Scuseria,
PRL 91, 146401 (2003)



Electron Localisation Function

$$\tau(\mathbf{r}) = \frac{1}{2} \sum_{i=1}^N |\nabla \psi_i(\mathbf{r})|^2 \quad \tau_W(\mathbf{r}) = \frac{|\nabla n(\mathbf{r})|^2}{8n(\mathbf{r})} \quad \tau_U(\mathbf{r}) \propto n(\mathbf{r})^{5/3}$$

Single-particle
kinetic energy density

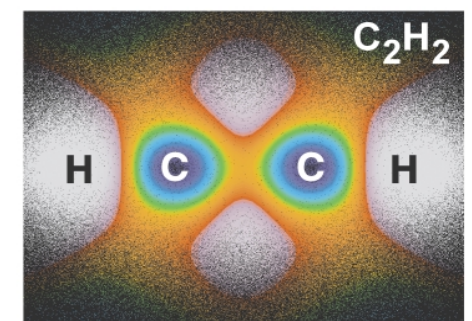
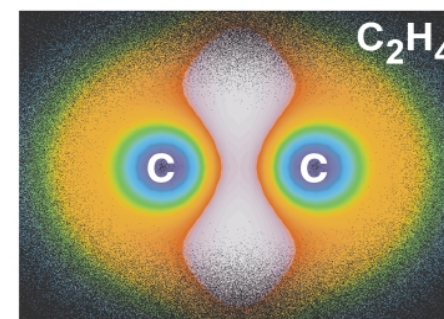
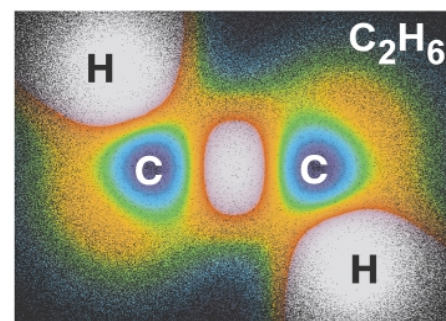
Weizsäcker
kinetic energy

uniform electron gas
kinetic energy density

$$\alpha(\mathbf{r}) = \frac{\tau(\mathbf{r}) - \tau_W(\mathbf{r})}{\tau_U(\mathbf{r})}$$

$$ELF(\mathbf{r}) = \frac{1}{1 + \alpha(\mathbf{r})^2}$$

$$0 \leq ELF(\mathbf{r}) \leq 1$$



- Detects bonding
 - single-orbital limit
 - uniform limit

• weak, non-covalent bonds

$$\begin{array}{lcl} \text{—————} & \alpha = 0 & \text{or } ELF = 1 \\ \text{—————} & \alpha = 1 & \text{or } ELF = 1/2 \\ \text{—————} & \alpha \gg 1 & \text{or } ELF = 0 \end{array}$$

mGGA parameterisations

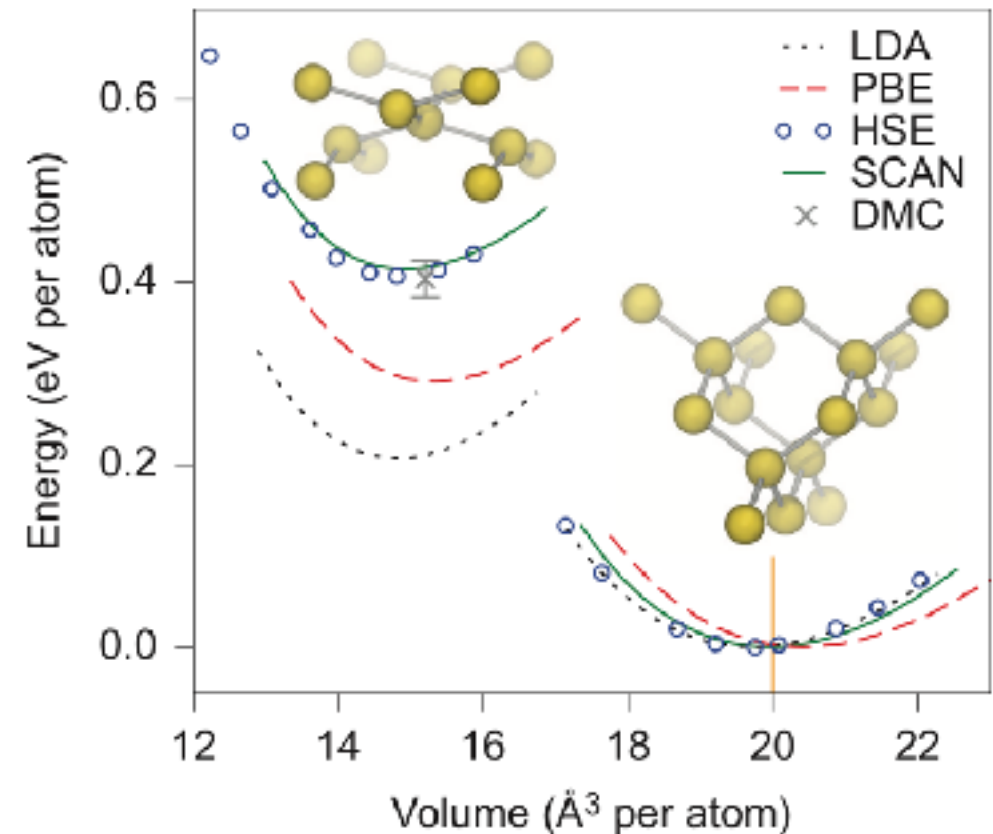
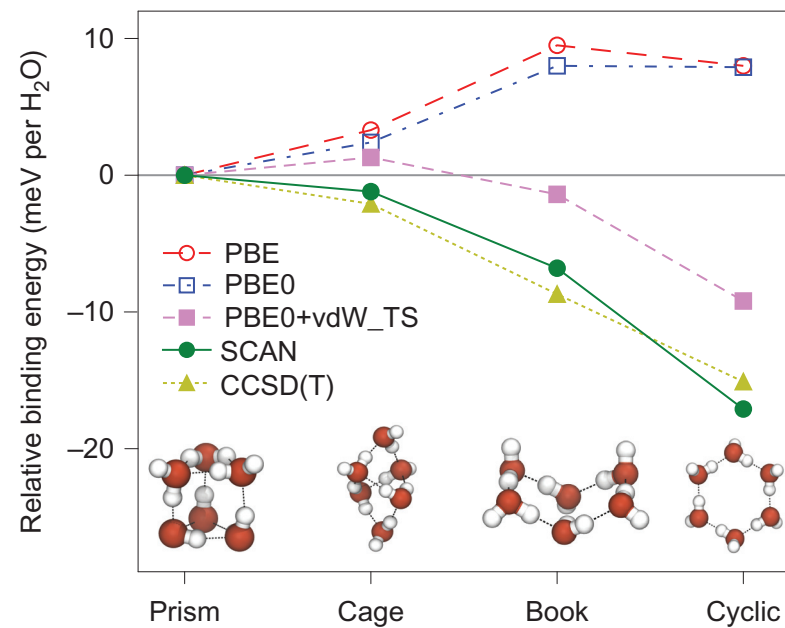
$$E_{xc} = \int d\mathbf{r} \, \varepsilon_{mGGA}(n(\mathbf{r}), \nabla n(\mathbf{r}), \nabla^2 n(\mathbf{r}), \tau(\mathbf{r})) n(\mathbf{r})$$

$$\tau(\mathbf{r}) = \frac{1}{2} \sum_{i=1}^N |\nabla \psi(\mathbf{r})|^2$$

$$V_{mBJ}(\mathbf{r}) = V(n(\mathbf{r}), |\nabla n(\mathbf{r})|, \tau(\mathbf{r}))$$

$$V_{mGGA}(\mathbf{r}) = \frac{\delta E_{xc}}{\delta n} = V_{GGA}(\mathbf{r}) + \hat{V}_{\tau}(\mathbf{r})$$

PRL **102**, 226401



Nat. Chem. **8**, 831

XC_FUNCTIONAL : rSCAN

- J. Sun, A. Ruzsinszky, J. P. Perdew

PRL **115** 036402 (2015)

- functional form obeys limits by construction
- bonding situation determined based on KED
- limiting cases are interpolated

CASTEP's libxc interface



- <https://libxc.gitlab.io/>
- over 700 (when I last checked) functionals
- native CASTEP
 - XC_FUNCTIONAL : LDA
 - XC_FUNCTIONAL : PBE
 - etc.
- libxc

```
%block xc_definition
LIBXC_GGA_X_PBE 1.0
LIBXC_GGA_C_PBE 1.0
%endblock xc_definition
```

- find definitions at <https://libxc.gitlab.io/functionals/>

GGA functionals

GGA exchange

- GGA_X_2D_B86 (id=128): Becke B6 in 2D
 - J. G. Vilhena, E. Rasanen, M. A. L. Marques, and S. Pittalis, J. Chem. Theory Comput. 10, 1837 (2014) (doi: 10.1021/ct400250g)
- GGA_X_2D_B86_MGC (id=124): Becke B6 with modified gradient correction for 2D
 - S. Pittalis, E. Rasanen, J. G. Vilhena, and M. A. L. Marques, Phys. Rev. A 79, 012503 (2009) (doi: 10.1103/PhysRevA.79.012503)
- GGA_X_2D_B88 (id=127): Becke B8 in 2D
 - J. G. Vilhena, E. Rasanen, M. A. L. Marques, and S. Pittalis, J. Chem. Theory Comput. 10, 1837 (2014) (doi: 10.1021/ct400250g)
- GGA_X_2D_PBE (id=129): Perdew, Burke & Ernzerhof in 2D
 - J. G. Vilhena, E. Rasanen, M. A. L. Marques, and S. Pittalis, J. Chem. Theory Comput. 10, 1837 (2014) (doi: 10.1021/ct400250g)
- GGA_X_AIRY (id=192): Constantin et al based on the Airy gas
 - L. A. Constantin, A. Ruzsinszky, and J. P. Perdew, Phys. Rev. B 80, 035125 (2009) (doi: 10.1103/PhysRevB.80.035125)
- GGA_X_AK13 (id=56): Armiento & Kuemmel 2013
 - R. Armiento and S. Kuemmel, Phys. Rev. Lett. 111, 036402 (2013) (doi: 10.1103/PhysRevLett.111.036402)
- GGA_X_AM05 (id=120): Armiento & Mattsson 05
 - R. Armiento and A. E. Mattsson, Phys. Rev. B 72, 085108 (2005) (doi: 10.1103/PhysRevB.72.085108)
 - A. E. Mattsson, R. Armiento, J. Paier, G. Kresse, J. M. Wills, and T. R. Mattsson, J. Chem. Phys. 128, 084714 (2008) (doi: 10.1063/1.289888)
- GGA_X_APBE (id=184): mu fixed from the semiclassical neutral atom

Hybrid functionals

$$E_{\text{x,exact}} = -\frac{1}{2} \sum_{i,j} \int d\mathbf{r} d\mathbf{r}' \frac{\psi_i^*(\mathbf{r})\psi_i(\mathbf{r}')\psi_j^*(\mathbf{r})\psi_j(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

$$E_{\text{x}} = (1 - \alpha)E_{\text{x,GGA}} + \alpha E_{\text{x,exact}}$$

$$V_{\text{x,exact}}(\mathbf{r}, \mathbf{r}') = -\frac{1}{2} \sum_j \frac{\psi_j^*(\mathbf{r})\psi_j(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

- Exact exchange: corresponds to the exchange energy of non-interacting one-body orbitals - does not account for all exchange energy
- Hybrid functionals use a mix of GGA exchange and exact exchange, as well as GGA correlation
- Non-local potential: a lot more expensive in plane-waves than semi-local functionals

Hybrid functional parameterisations

In CASTEP:

- Worse scaling with system size - more expensive
- Not all functionality present

XC_FUNCTIONAL : B3LYP

- P. J. Stephens, F. J. Devlin, C. F. Chabalowski and M. J. Frisch, *JPC* **98** 11623 (1994)

XC_FUNCTIONAL : PBE0

- C. Adamo and V. Barone, *JCP* **110** 6158 (1999)

range-separated
hybrids (more
efficient for metals)

XC_FUNCTIONAL : HSE03

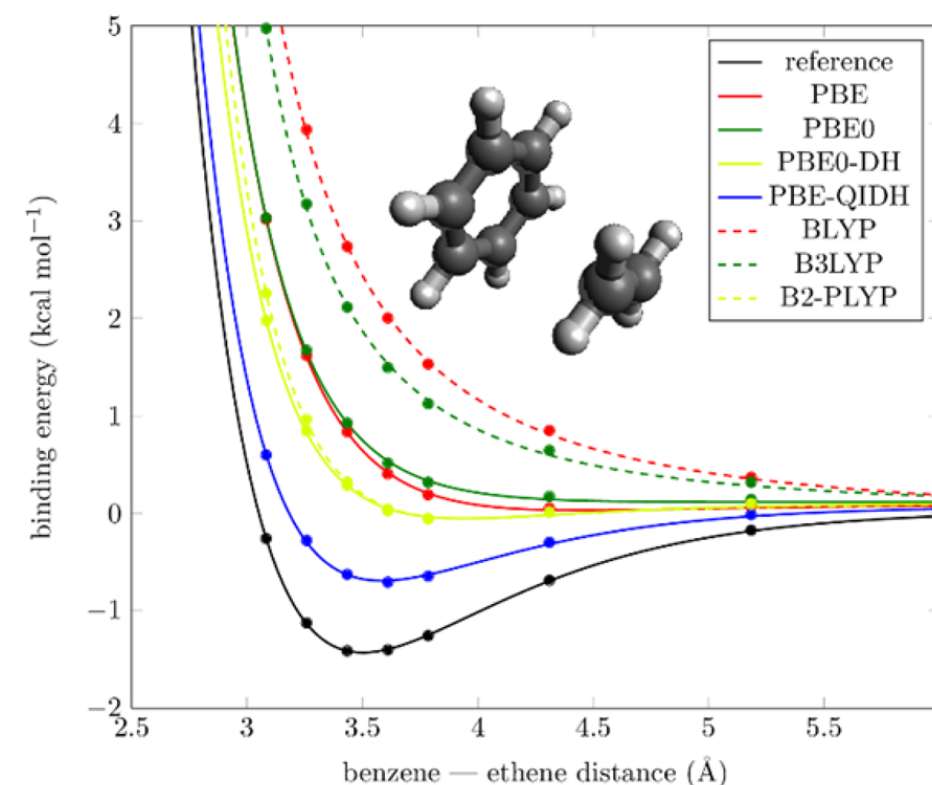
- J. Heyd, G. E. Scuseria and M. Ernzerhof, *JCP* **118** 8207 (2003)

XC_FUNCTIONAL : HSE06

- A. V. Krukau, O. A. Vydrov, A. F. Izmaylov, and G. E. Scuseria, *JCP* **125** 224106 (2006)

Double hybrid functionals

- Hybrid exchange and
- In a similar spirit to hybrids, correlation is a mix between GGA correlation and MP2 correlation
- Weak interactions (van der Waals) better described
- Not available in CASTEP

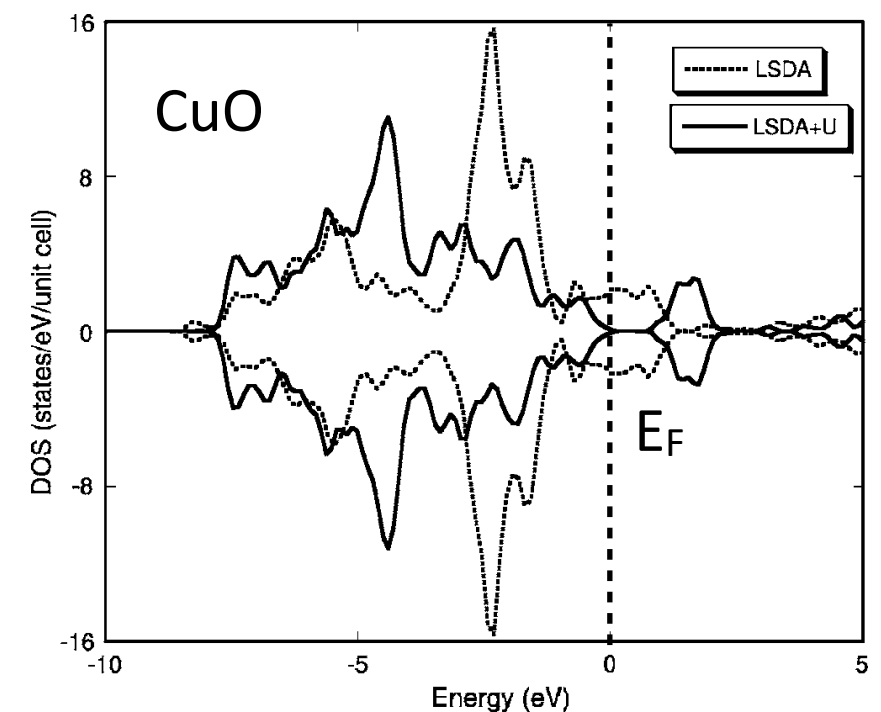
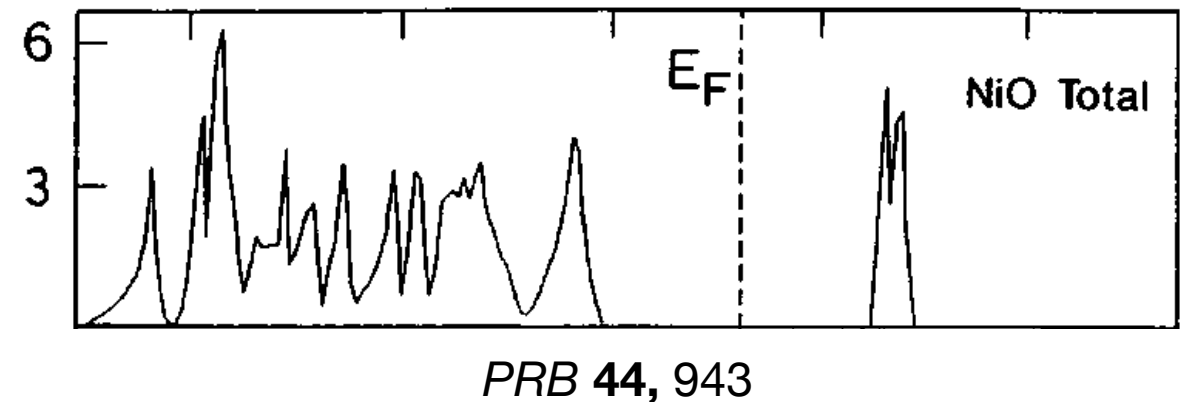


Acc. Chem. Res. **49**, 1503

DFT+U

- Self-interaction error in DFT
- On-site repulsion not treated properly
- Correlation effects not modelled accurately especially in open-shell systems
- Electrons in LDA and GGA tend to be more delocalised than they should be
- meta-GGAs (SCAN) appears to improve the description of strongly correlated systems *without* U
- U becomes an adjustable parameter... or
- CASTEP can determine U from linear response

```
%block hubbard_u  
Ni d:5.0  
%endblock hubbard_u
```



Dispersion corrections

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- Not all correlation effects accounted for in DFT (particularly non-local correlation)
- van der Waals interactions significant in a number of systems:
 - Molecular crystals
 - Layered materials
 - Even oxides!
- Dispersion correction:
 - Pairwise schemes
 - Three-body interactions
 - TS - pairwise parameters depend on electronic structure
 - many-body dispersion



JCP **140**, 18A508

`sedc_apply : true`

`sedc_scheme : OBS`

• PRB 73, 205101 (2006)

`sedc_scheme : G06`

• (D2) J. Comput. Chem. 27, 1787 (2006)

`sedc_scheme : JCHS`

• J. Comput. Chem. 28, 555 (2007)

`sedc_scheme : TS`

• PRL 102, 073005 (2009)

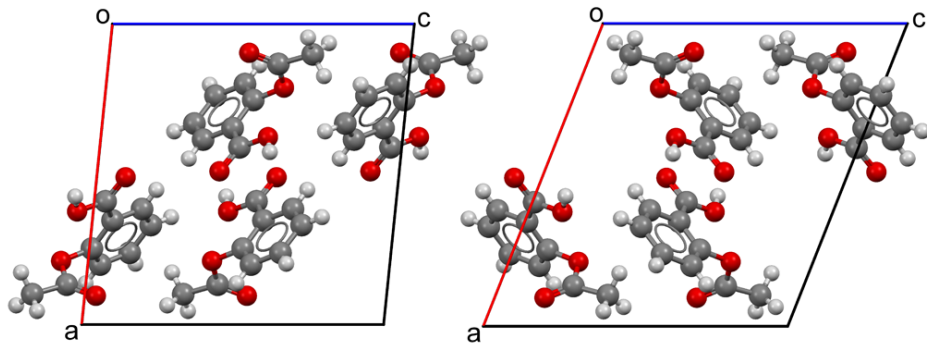
`sedc_scheme : MBD*`

• J. Chem. Phys. 140, 18A508 (2014)



Dispersion corrections

Two forms of aspirin

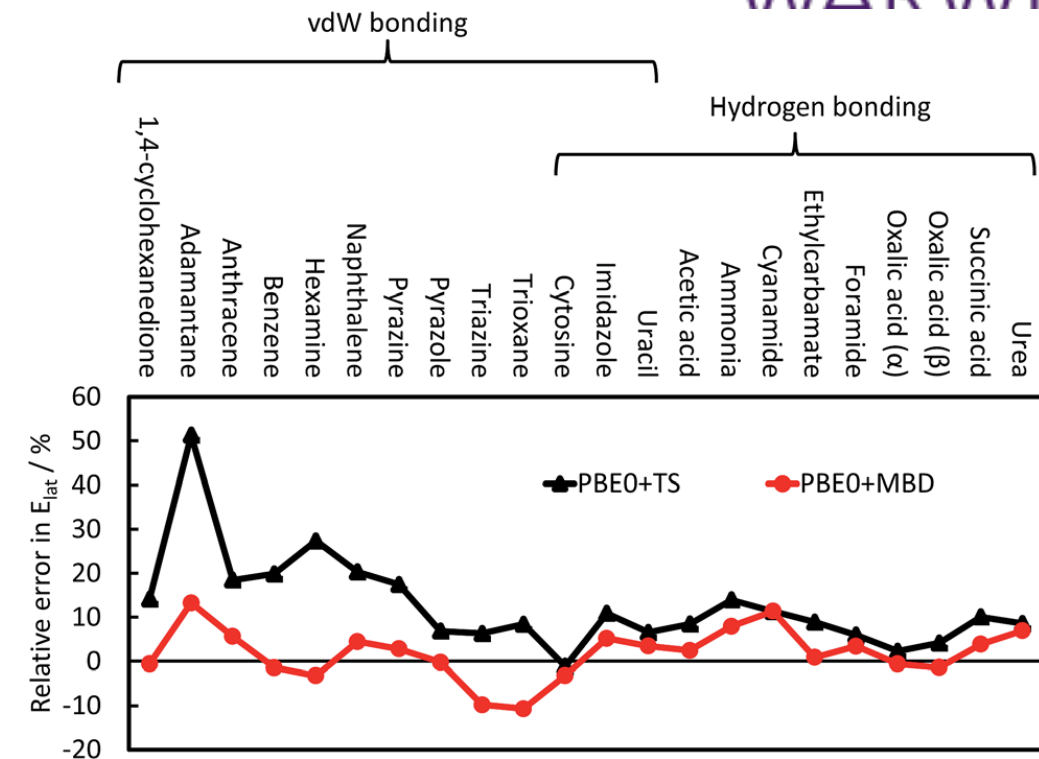


	$\Delta E_{I \rightarrow II} / \text{kJ/mol}$
PBE + TS	-0.18
PBE + TS + ZPE	-0.42
PBE + TS + $F_{\text{vib}}(298 \text{ K})$	-0.68
PBE + MBD	0.04
PBE + MBD + ZPE	0.35
PBE + MBD + $F_{\text{vib}}(298 \text{ K})$	2.56

PRL **113**, 055701

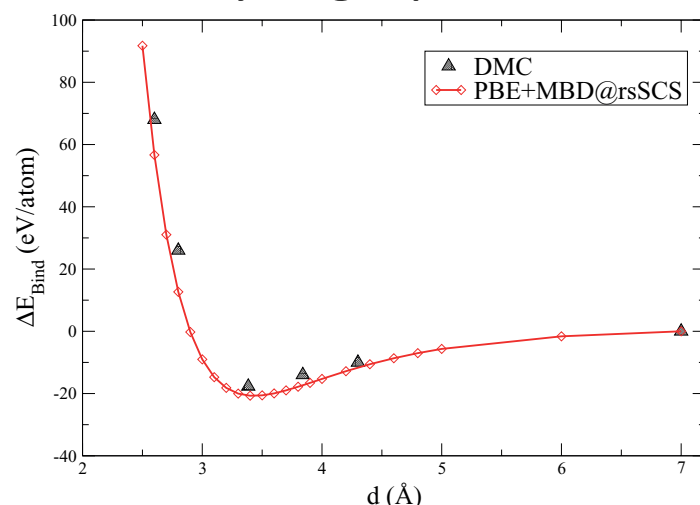
Molecular crystals

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Chem. Sci., **6**, 3289

Bilayer graphene



J. Phys.: Condens. Matter **28** 045201

Graphite and BN

Method	a_0 (Å)	c_0 (Å)	E_B (meV atom ⁻¹)	B_0 (GPa)	Reference
Graphite (AB stacking)					
Expt.	2.46	6.71	43, 35 ± 10, 52 ± 5	34–42	[27, 46–48]
RPA	n.a.	6.68	48	36	[34]
PBE + MBD@ rsSCS	2.46	6.82	48	29	This work
Graphite (AA stacking)					
RPA	n.a.	3.5	36	18	
PBE + MBD@ rsSCS	2.46	3.61	40	26	This work
h-BN					
Expt.	2.50	6.66	—	37	[49]
RPA	n.a.	6.60	39	n.a.	[35, 50]
PBE + MBD@ rsSCS	2.50	6.59	59	30	This work

Which functional to use?

- LDA
 - nice covalent systems
 - simple metals
- DFT+U
 - strongly correlated systems (localised d and f orbitals)
 - Mott insulators
- GGA
 - molecules
 - hydrogen-bonded materials
 - transition metals
 - magnetic systems
- Hybrid and screened hybrid exchange
 - band gaps
 - magnetic systems
- Dispersion corrected DFT (mostly GGA)
 - molecular crystals
 - layered materials
- meta-GGA
 - careful optimism: improves on most properties, including mid-range van der Waals